Sarthak Mishra

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Summary

CS Final Year focused on security engineering, application security, and vulnerability research. Experienced in building security tools and contributing Several Merged PRs to SigmaHQ and ProjectDiscovery's Open-Source Repositories.

Technical Skills

Application & Offensive Security: OWASP Top 10, Web Application Security, API Security, Authentication & Authorization, Vulnerability Assessment, Vulnerability Validation, Burp Suite, Nuclei

Security Engineering & Detection: Threat Detection, Detection Engineering, Log Analysis, Event Correlation, IOC Analysis, MITRE ATT&CK, Sigma Rules, SIEM

Security Tools: Splunk, SPL, Microsoft Sentinel, KQL, Wireshark, Nmap, Scapy, Git

Programming & Scripting: Python, Go, SQL, Bash, C++

Networking & Systems: TCP/IP, DNS, HTTP/HTTPS, TLS, Linux, Windows, Network Analysis, Packet Analysis

Backend & Infrastructure: FastAPI, Flask, REST APIs, PostgreSQL, GitHub Actions

Open-Source Contributions

ProjectDiscovery – Nuclei Templates

- Improved vulnerability detection across UNA CMS RCE, Azure Pipelines, SAP ReadConfigFile, OneNav SSRF, and CVE-2026-31807 by fixing false-positive conditions and refining detection logic.
- Implemented conditional HTTP requests, strict response matching, content-type filtering, and dynamic payload validation; fixed Dell DPI false negatives, CVE-2025-12536 metadata, and documented ECDSA template signing and digest integrity workflows.

SigmaHQ – Sigma

- Optimized automated testing by eliminating repeated YAML parsing and reusing loaded rule data, achieving a 5.4× reduction in CI execution time (approximately 75 seconds).
- Fixed Windows process detection across seven Sigma rules through `OriginalFileName` matching corrections, and improved Microsoft Office persistence detection by correcting executable path and registry key filters.

Projects

Xentinel – SOC Telemetry & Detection Platform *| Python, Flask, Linux, JSON*

- Built a real-time SOC simulation platform that generates authentication telemetry, processes security events, and produces severity-based alerts through a modular detection pipeline.
- Engineered detections for SSH brute-force, password-spray, user-enumeration, suspicious administrative logins, and multiple-session abuse, with MITRE ATT&CK technique mapping and source-IP telemetry.

Arpex – ARP Spoofing Detection & Network Monitoring *| Python, Scapy, Flask, SQLite, Pytest*

- Engineered a real-time ARP monitoring system that discovers network devices, establishes IP-to-MAC baselines, and detects suspicious mapping changes.
- Implemented verification workflows to validate anomalous ARP activity and reduce false positives, while tracking attacks, affected devices, and security events.

Hexora – Static File Analysis & Triage Platform *| Python, FastAPI, SQLAlchemy, SQLite*

- Built a static file-analysis platform that inspects files without execution, using cryptographic hashing, file-signature detection, metadata extraction, entropy analysis, and heuristic risk scoring for security triage.
- Implemented detection of suspicious strings, URLs, IP addresses, and file characteristics to identify artifacts requiring further investigation.

Certifications

- **Certified Ethical Hacker (CEH) v13** – EC-Council *(Training in Progress; Exam Enrolled)*
- **Cyber Security 101** – TryHackMe
- **Pre Security** – TryHackMe

Education

Vellore Institute of Technology *Bhopal, India*
Bachelor of Technology in Computer Science and Engineering **CGPA:** 7.70/10.00